

Vanishing Integral Homology for Non-Uniquely Run-Embedded Binary Insertion Intervals with $|v| \leq 6$

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Abstract

The insertion chain complex of a closed subsequence interval is known to have vanishing positive-degree integral homology under a unique run-exponent-vector hypothesis. We give an exact exhaustive result on the complementary cases at short binary length. For every binary pair $u \preceq v$ with $|v| \leq 6$ for which the run-exponent vector is not unique, the complete interval $[u, v]$ has $H_j = 0$ over $\mathbb{Z}$ for all $j \geq 1$. There are exactly 424 such intervals. Complete canonical cells and signed boundaries were regenerated in two ways, and 2,544 integral filling certificates were checked by exact arithmetic. This is a finite computer-assisted theorem, not a structural contraction: the unrestricted conjecture for arbitrary alphabets and lengths is not resolved here, and no claim of global priority is made.

1 Statement and context

For words over a finite alphabet, write $u \preceq v$ when u is a not-necessarily-contiguous subsequence of v , and put

$$[u, v] = \{w : u \preceq w \preceq v\}.$$

Jonoska, Martinez-Figueroa, and Saito introduced an insertion chain complex for word sets and proved positive-degree integral homology vanishing for these closed intervals under a uniqueness condition [1].

To state that condition, write the maximal constant-letter run decomposition as

$$v = c_1^{r_1} c_2^{r_2} \cdots c_s^{r_s}, \quad c_i \neq c_{i+1}, \quad r_i \geq 1.$$

A *run-exponent vector* for u relative to v is a vector $q = (q_1, \dots, q_s)$ with $0 \leq q_i \leq r_i$ such that $u = c_1^{q_1} \cdots c_s^{q_s}$, with zero powers omitted. Every $u \preceq v$ has at least one such vector. The cited theorem applies when it has exactly one.

Our contribution treats every short binary interval outside that hypothesis.

Theorem 1 (Exact short-binary complement). *Let $\Sigma = \{a, b\}$. For every $u \preceq v$ with $|v| \leq 6$ such that u has more than one run-exponent vector relative to the maximal constant-letter runs of v , the insertion-block chain complex of the complete closed interval $[u, v]$, with the alternating integral boundary defined below, satisfies*

$$H_j(C_*([u, v]; \mathbb{Z})) = 0 \quad \text{for every } j \geq 1.$$

There are exactly 424 pairs in this quantified family.

Combining Theorem 1 with the unique-vector theorem in [1] gives the following convenient reformulation.

Corollary 2. *If a binary closed subsequence interval is a counterexample to universal positive-degree integral homology vanishing, then its top word has length at least 7.*

The restriction in Theorem 1 is essential to the present result: we do not give a uniform contraction, and we make no assertion about longer binary words or alphabets of size at least three.

2 The complex and the finite universe

An m -insertion block has the form

$$x_0(\varepsilon, a_1)x_1 \cdots (\varepsilon, a_m)x_m, \quad x_i \in \Sigma^*, \quad a_i \in \Sigma.$$

Independently replacing each (ε, a_i) by ε or a_i gives its indexed vertex words. Repeated vertex words are permitted. We use the canonical form in which, for $i \geq 1$, a nonempty x_i does not begin with a_i ; it is obtained by repeatedly applying $(\varepsilon, a)a \mapsto a(\varepsilon, a)$. A canonical block is valid precisely when

$$x_i = \varepsilon \implies a_i \neq a_{i+1} \quad (1 \leq i < m).$$

The group $C_m(W; \mathbb{Z})$ is freely generated by the valid canonical m -blocks whose indexed vertices all lie in W .

Let F_i^+ fix the i th optional letter as a_i , and let F_i^- delete it. After either operation the face is normalized, and an invalid lower face is set to zero. The boundary convention used throughout is

$$\partial_m = \sum_{i=1}^m (-1)^{i+1} (F_i^+ - F_i^-). \quad (1)$$

This is the alternating convention used in the appendix proof of [1]. All matrix products $\partial_{m-1}\partial_m$ were also checked entrywise over $\mathbb{Z}$ in the finite computation.

The following elementary parametrization makes exhaustive cell generation small and transparent.

Lemma 3 (Run parametrization of cells). *Fix the maximum vertex word w of a canonical valid block. Its m optional letters are exactly the final letters of m distinct maximal constant-letter runs of w . Consequently, the canonical valid m -blocks with maximum w are in bijection with the m -subsets of the runs of w ; support in $[u, v]$ is decided by testing all 2^m indexed vertices.*

Proof. Within a constant-letter run, canonicity moves every fixed copy before an optional copy. Two optional copies in that same run violate validity after normalization. Thus at most one letter in a run is optional, and, when it is, canonicity puts it last. Conversely, marking the final letter in each of a set of distinct runs gives a canonical block; adjacent marked runs have different letters, so the validity condition holds. The definition of $C_m([u, v]; \mathbb{Z})$ then leaves exactly the stated vertex-support test. No injectivity condition is imposed on the map from choices to vertex words. $\square$

The quantified family was generated as follows.

1. Enumerate every $v \in \{a, b\}^{\leq 6}$ and the set of all distinct deletion-subsequences u of v .
2. Enumerate the finite Cartesian box $0 \leq q_i \leq r_i$ and retain a pair exactly when more than one vector spells u .

3. Generate the complete interval by retaining every subsequence w of v with $u \preceq w$, and use Lemma 3 in every degree.
4. Form every face in (1), normalize it, combine coincident faces with their integer signs, and record the matrices D_j of ∂_j .

Table 1 gives the complete universe count. In particular, the 424 cases are not a sample selected after a homology calculation.

Table 1: Exhaustive binary pair counts by top-word length.

$ v $	distinct pairs $u \preceq v$	pairs with multiple run vectors
0	1	0
1	4	0
2	14	0
3	46	2
4	146	16
5	454	80
6	1394	326
Total	2059	424

3 Integral certificates

We record the certificate logic so that vanishing does not rest on a software homology command or on ranks over a field.

Lemma 4 (Degree-one filling certificate). *Let D_1 be a signed graph-incidence matrix. Choose a spanning forest and regard all other edges, including zero-incidence loops, as chords. Let the columns of K_1 be the fundamental integral cycles, oriented so that chord projection is the identity. If an integer matrix R_1 satisfies*

$$D_2 R_1 = K_1,$$

then $\ker_{\mathbb{Z}} D_1 = \text{im}_{\mathbb{Z}} D_2$.

Proof. Deleting chord coordinates from an integral cycle leaves a cycle supported on a forest, hence zero. Conversely, completing each chord along its unique forest path (and taking a loop by itself) gives an integral cycle. Chord projection therefore identifies $\ker_{\mathbb{Z}} D_1$ with a free integer lattice having basis K_1 . The displayed identity fills every basis cycle. $\square$

For higher degree we use a short lattice certificate.

Lemma 5 (Primitive-kernel filling certificate). *Let $D : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ and $B : \mathbb{Z}^\ell \rightarrow \mathbb{Z}^n$ satisfy $DB = 0$. Suppose integer matrices $K \in \mathbb{Z}^{n \times k}$, $P \in \mathbb{Z}^{k \times n}$, and $R \in \mathbb{Z}^{\ell \times k}$ satisfy*

$$DK = 0, \quad PK = I_k, \quad BR = K. \tag{2}$$

If, for some prime p ,

$$\text{rank}_{\mathbb{F}_p}(D \bmod p) = n - k,$$

then $\ker_{\mathbb{Z}} D = \text{im}_{\mathbb{Z}} B$.

Proof. The first two identities make the k columns of K independent elements of $\ker D$. Hence $\text{rank}_{\mathbb{Q}} D \leq n - k$. Rank cannot decrease when the modular lower bound is lifted to $\mathbb{Q}$, so the assumed equality gives $\dim_{\mathbb{Q}} \ker D = k$. Every $x \in \ker_{\mathbb{Z}} D$ can therefore be written $x = Kq$ over $\mathbb{Q}$. Applying P gives $q = Px \in \mathbb{Z}^k$, so the columns of K form a basis of the full integral kernel, not merely a finite-index sublattice. The third identity in (2) fills that basis. $\square$

For each interval, D_1 was checked columnwise to be a signed incidence matrix. A deterministic forest/chord partition produced K_1 . If A_1 is the chord-row submatrix of D_2 , the exact identities

$$D_2 = K_1 A_1, \quad A_1 R_1 = I, \quad D_2 R_1 = K_1$$

were verified. Lemma 4 proves $H_1 = 0$ over $\mathbb{Z}$ in all 424 cases.

For every $2 \leq j \leq 6$, exported matrices K_j, P_j, R_j were checked against

$$D_j K_j = 0, \quad P_j K_j = I, \quad D_{j+1} R_j = K_j.$$

Exact Gaussian elimination modulo 2 supplied the rank equality in Lemma 5. Thus every integral j -cycle is an integral boundary. This accounts for 424 degree-one and $5 \cdot 424 = 2,120$ higher-degree certificate checks. Finally, a cell cannot have more optional letters than the length of its maximum word, so $C_j = 0$ for $j > 6$.

The largest groups and kernels encountered are shown in Table 2. Empty matrices and zero-rank cases were retained in the quantified verification.

Table 2: Maximum ranks among the 424 complexes.

j	0	1	2	3	4	5	6
$\max \text{rank } C_j$	29	76	86	54	19	3	0
$\max \text{rank } \ker D_j$	—	48	38	16	3	0	0

4 Reproducibility and checks

The accompanying data retain the complete cell lists, signed matrices, and integer certificates, rather than only the zero-homology summary. One implementation generates cells directly from maximal runs as in Lemma 3. A separately written checker starts from arbitrary optional positions, applies the local swap normalization, discards invalid blocks, and regenerates every $C_0, \dots, C_6$ and $D_1, \dots, D_6$. It also reenumerates all 127 top words, all 2,059 distinct pairs, and the 424 filtered pairs. The two cell sets and boundary matrices agree after reindexing, and every adjacent boundary product is zero over $\mathbb{Z}$.

Certificate generation used exact Smith transformations, but verification of the released conclusion does not trust that implementation: it checks the displayed integer identities entrywise, reconstructs degree-one kernel cycles by tree elimination, and uses an independent modular rank computation in higher degrees. The computation used Python 3.9.6 and SymPy 1.14.0. The principal machine-readable artifacts are `research-pass1.json`, `research-pass2.json`, `research-pass3.json`, their corresponding programs and cross-checks, and `referee-verify.py`. Re-running the verifier from these exact finite data is sufficient to check every certificate asserted above.

5 Scope of the literature comparison

The mathematical comparison made here is deliberately narrow. Reference [1] supplies the definition, the unique-run-vector theorem, and the unrestricted conjectural direction that motivates Theorem 1. The present computation covers the finite complement through binary top length six; it is neither an application of the cited theorem nor a proof of the unrestricted statement.

Exact enumeration followed by independently replayable integer certificates is a broader pattern in computer-assisted proofs. For a methodologically adjacent enumerative example, unrelated to insertion complexes or their homology, see [2]. That reference is included only as broad computational context and supplies no support for Theorem 1. Beyond the defining 2025 source, a current comprehensive literature and priority review was not established in this bounded work. Accordingly, this note makes no claim to be first, optimal, or exhaustive in the literature.

Computational and AI assistance disclosure. Python and SymPy were used for exact enumeration, integer linear algebra, and certificate serialization. OpenAI Codex assisted with program development, cross-check design, and preparation of this manuscript. The mathematical conclusion is based on the explicit finite enumeration and exact certificate identities described above; generated prose or model confidence was not treated as mathematical evidence.

References

- [1] Nataša Jonoska, Francisco Martinez-Figueroa, and Masahico Saito, *The Insertion Chain Complex: A Topological Approach to the Structure of Word Sets*, arXiv:2509.12607v1 (2025). <https://arxiv.org/abs/2509.12607>.
- [2] Zijian Zeng, *The Distinct-Cycle-Length Probability is Strictly Decreasing after 30*, SSRN working paper 7380519 (2026). <https://doi.org/10.2139/ssrn.7380519>.